

USER MANUAL

AuraTune

A step-by-step guide to running AuraTune's adaptive EQ dashboard, reading its results, and setting up your real headphone or speaker app to match.

Version covers the local ML genre-aware build · Guide generated 2026-09-16

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1. What AuraTune does

AuraTune listens to two things — the noise in your room and what you're playing (a podcast, music, or a movie) — and works out how your equalizer should change to sound its best. It explains every change in one plain-English sentence, and if you tell it which real EQ app or headphones you use, it tells you the **exact slider values** to set.

No setup required

Runs entirely with built-in demo scenarios and sensible defaults — nothing to configure before your first run.

Explains itself

Every adjustment comes with a plain-English reason, not just numbers.

Matches your real gear

Snaps its recommendation onto your actual EQ app's sliders, step size, and range.

2. Starting the app

From a terminal, inside the project folder:

```
streamlit run app.py
```

This opens the dashboard in your browser at <http://localhost:8501>. You'll land on a clean, empty state like this:

Deploy

1. ADAPTIVE AUDIO PERSONALIZATION

AuraTune

Perception → 3-agent LangGraph → your EQ, explained in plain English

Scenario

Simulated context

Quiet room + podcast

Live command (optional)

e.g. make voices clearer, less bass

Genre model (local ML)

auto (best)

Your EQ app

Which EQ are you dialing in?

Wavelet - Graphic EQ (9-band)

9 bands · -12...+12 dB · 0.1 dB step

Read from the user's screenshot (WhatsApp image 2026-09-10 4:37)

Ready when you are

Pick a scenario, choose your EQ app on the left, and click Run adaptation to see the live curve and explanation appear here.

Everything works out of the box with no accounts or API keys. Setting `ANTHROPIC_API_KEY` or `GEMINI_API_KEY` later unlocks two extra things (free-text command understanding via Claude, and reading a screenshot of your EQ app) — see section 5.

3. Step-by-step walkthrough

1 Pick a Scenario

Choose which situation you want AuraTune to react to: **Quiet room + podcast**, **Noisy environment + music**, or **Home + movie**. This stands in for a live microphone and whatever's actually playing.

2 Type a live command (optional)

Tell it what you want in plain English — e.g. *"make voices clearer, less bass"*. AuraTune folds this directly into the EQ curve on top of whatever the room/content already called for.

3 Pick a genre model (optional)

If a local ML model has been trained (see section 6), you can choose which one classifies the music's genre/mood — or leave it on **auto (best)**.

The screenshot shows the AuraTune web interface. At the top right is a 'Deploy' button. The main heading is 'AuraTune' with a subtitle 'Perception → 3-agent LangGraph → your EQ, explained in plain English'. On the left is a configuration panel with three sections: 'Scenario' with a 'Simulated context' dropdown set to 'Noisy environment + music', a 'Live command (optional)' text input with 'make voices clearer, less bass' and a 'Press Enter to apply' button, and a 'Genre model (local ML)' dropdown set to 'auto (best)'. Below this is 'Your EQ app' section with a 'Which EQ are you dialing in?' dropdown set to 'Wavelet - Graphic EQ (9-band)' and technical details '9 bands · -12...+12 dB · 0.1 dB step'. At the bottom of this panel is a small text 'Read from the user's screenshot (WhatsApp image 2026-09-10 4.37)'. On the right is a 'Ready when you are' box with a lightbulb icon and text: 'Pick a scenario, choose your EQ app on the left, and click Run adaptation to see the live curve and explanation appear here.'

4 Tell it your EQ app

Under **Your EQ app**, choose a built-in preset, upload a screenshot of your actual app, or type one in by hand. See section 5 for the three ways to do this.

5 Click Run adaptation

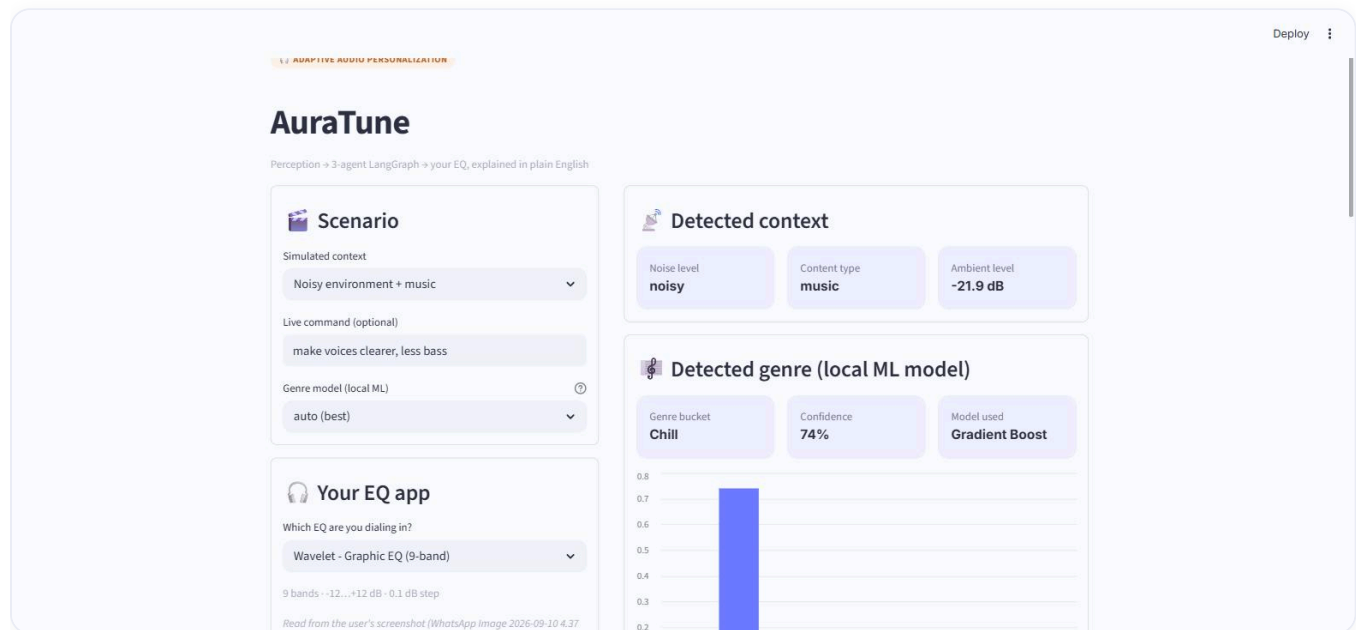
AuraTune runs its full pipeline — perception, the agent pipeline, and the DSP engine — and fills the right-hand side with results, usually in under a second.

4. Reading your results

After clicking **Run adaptation**, you'll see (top to bottom):

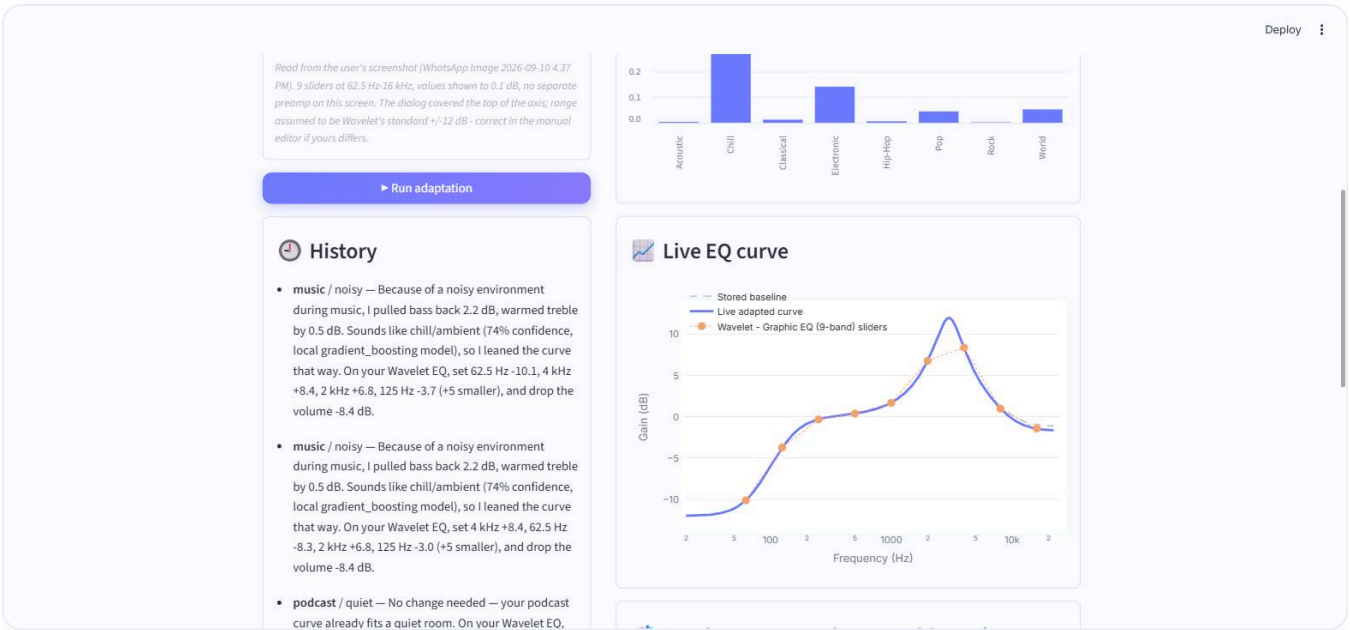
Detected context & detected genre

What AuraTune picked up from the room and the content — noise level, content type, ambient loudness, and (for music) the genre/mood bucket its local ML model predicted, with a confidence score and a bar chart of every genre it considered.



Live EQ curve

A chart comparing your **stored baseline** (dashed grey — what you normally use for this content type) against the **live adapted curve** (solid, the new recommendation) and, if you've set an EQ app, the exact points on **your app's sliders** (dotted markers).



5. Setting up "Your EQ app"

Every real EQ is a fixed grid of sliders — AuraTune needs to know yours to give you exact numbers instead of just a smooth curve. Three ways to set it, from the dropdown:

Method	What to do	Needs
 Upload a screenshot	Snap your EQ app's screen and upload it. AuraTune reads the band frequencies, step size, and range automatically.	A free Google Gemini key (30 seconds to get one) or an Anthropic key
Presets	Pick from built-in specs — Wavelet, a generic 10-band ISO EQ, Spotify, a car head unit, plus any your team has added (Sony, Samsung, Bose, Sennheiser, Soundcore, ...).	Nothing
Manual...	Type in the band list, step size, and range by hand. You can save it for next time.	Nothing

Pick **(none — just show the curve)** if you'd rather see the abstract curve without slider-specific numbers.

6. The local ML genre model

For music content, AuraTune can classify the genre/mood (electronic, rock, hip-hop, pop, acoustic, classical, chill, or world/latin) using a model trained locally on a 114,000-track real dataset — no internet connection needed to use it. If you see *"No trained genre model found"* instead of the genre picker, someone needs to run the one-time training step:

```
python ml/train.py
```

Takes a couple of minutes. Once done, refresh the app and the "Genre model (local ML)" picker and results panel appear automatically.

7. History & repeat use

The **History** panel on the left keeps a running log of every adaptation you've run, most recent last, so you can compare how different scenarios and commands were handled. Each run also quietly updates your stored baseline for that content type, so future runs start from your latest preference.

8. Troubleshooting & tips

What you see	What it means
Explanation sounds a bit templated/plain	No <code>ANTHROPIC_API_KEY</code> is set, so AuraTune is using its built-in rule-based sentence instead of Claude's phrasing. Everything still works — it's just less conversational.

"No trained genre model found"	Run <code>python ml/train.py</code> once (see section 6). Until then, genre-aware tuning is simply skipped — nothing breaks.
Screenshot upload says it needs a key	Paste a free Gemini key into the field that appears, or set <code>GEMINI_API_KEY</code> / <code>ANTHROPIC_API_KEY</code> in your environment.
"Some bands hit the app's range limit"	The ideal curve asked for more boost/cut than your real EQ app allows — AuraTune gave you the closest value it can reach.
Movie/podcast scenarios don't show a genre panel	Expected — genre classification only runs for music content.